

A full answer to the odd-dimensional case in arXiv:1905.12257

Candidate solution packet for `fa_banach_001`

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Source and Target

The source paper is Kadets, Martin, Meri, Perez, and Quero, *On the numerical index with respect to an operator*, arXiv:1905.12257 [1]. After Theorem 4.7 the authors prove the desired zero-value result for $\mathcal{L}(H)$ and $\mathcal{K}(H)$ when the real Hilbert space H is infinite-dimensional or even-dimensional, and then leave the finite odd case open.

index zero is given by

$$J[(x_\lambda)_{\lambda \in \Lambda}] = (Ax_\lambda)_{\lambda \in \Lambda} \quad ((x_\lambda)_{\lambda \in \Lambda} \in H).$$

Now, the operator Φ_J is an onto isometry on $\mathcal{L}(H)$ or $\mathcal{K}(H)$ ($\Phi_{J^{-1}}$ is clearly the inverse of Φ_J) satisfying that $v(\Phi_J) = 0$. Then, Corollary 3.11 gives the result. $\square$

When H has odd dimension, we do not know if the equality $n_G(\mathcal{L}(H), Y) = 0$ holds for every Banach space Y and every operator $G \in \mathcal{L}(\mathcal{L}(H), Y)$.

Another result of the same kind tells us that there are many other spaces of operators having trivial set of values of the numerical indices with respect to an operator.

The question in context is whether, for a finite-dimensional real Hilbert space H of odd dimension at least 3,

$$n_G(\mathcal{L}(H), Y) = 0$$

holds for every Banach space Y and every norm-one operator $G \in \mathcal{L}(\mathcal{L}(H), Y)$. Dimension 1 is outside the context of Theorem 4.7 and would be exceptional.

Idea of the Proof

The even-dimensional proof in the source paper uses a surjective isometry of H with numerical radius zero, then transports it by left multiplication to $\mathcal{L}(H)$. Such an invertible skew isometry cannot exist in odd real dimension. The replacement is to use two-sided infinitesimal isometries on $\mathcal{L}(H)$:

$$X \longmapsto AX + XB,$$

where A and B are skew-adjoint on H . These maps have numerical radius zero because they generate the isometry group $X \mapsto e^{tA} X e^{tB}$. By choosing the one-dimensional kernels of A and B to match a single Hilbert-Schmidt orthogonality condition, every prescribed operator $C \in \mathcal{L}(H)$ lies in the range of one such zero-radius map. This gives exactly the range-covering hypothesis in Proposition 3.10(c) of the source paper.

Theorem

Theorem 1. *Let H be a real Hilbert space of odd dimension $n \geq 3$, and put $W = \mathcal{L}(H)$ with its operator norm. Then*

$$n_G(W, Y) = 0$$

for every Banach space Y and every norm-one operator $G \in \mathcal{L}(W, Y)$. Consequently the question after Theorem 4.7 of [1] has an affirmative answer.

Proof. By Proposition 3.10(c) of [1], it is enough to prove that

$$\bigcup \{R(W) : R \in \mathcal{L}(W), v(R) = 0\}$$

is dense in W . We prove the stronger statement that this union is all of W .

Fix $C \in W$. Choose a unit vector $b \in H$. Since $n \geq 3$, we may choose a unit vector $a \in H$ such that

$$\langle Cb, a \rangle = 0;$$

if $Cb = 0$, choose any unit a . Write $n = 2m + 1$.

Choose skew-adjoint operators $A, B \in \mathcal{L}(H)$ such that

$$\ker A = \mathbb{R}a, \quad \ker B = \mathbb{R}b,$$

and such that the nonzero rotation frequencies of A and B are disjoint. Concretely, decompose $H = \mathbb{R}a \oplus E_A$ and $H = \mathbb{R}b \oplus E_B$, split each even-dimensional complement into orthogonal 2-planes, let A be block rotation by positive frequencies $\alpha_1, \dots, \alpha_m$ on E_A , let B be block rotation by positive frequencies $\beta_1, \dots, \beta_m$ on E_B , and choose the two finite sets $\{\alpha_i\}$ and $\{\beta_j\}$ disjoint.

Define the Sylvester operator $R \in \mathcal{L}(W)$ by

$$R(X) = AX + XB \quad (X \in W).$$

First, $v(R) = 0$. Indeed, because A and B are skew-adjoint, e^{tA} and e^{tB} are orthogonal for every real t , and hence

$$U_t(X) = e^{tA} X e^{tB}$$

is a surjective isometry of W . The left and right multiplication parts commute, so $U_t = e^{tR}$. The derivative formula for the numerical range used in Lemma 2.4 of [1] gives

$$\sup \operatorname{Re} V(\mathcal{L}(W), \operatorname{Id}_W, R) = \lim_{t \downarrow 0} \frac{\|\operatorname{Id}_W + tR\| - 1}{t}.$$

Since e^{tR} is an isometry and $\operatorname{Id}_W + tR = e^{tR} + O(t^2)$ in operator norm, the right-hand side is at most 0. Applying the same argument to $-R$ shows that the real numerical range of R is contained in $\{0\}$. Therefore $v(R) = 0$.

It remains to show that $C \in R(W)$. Use the Hilbert-Schmidt inner product on the finite-dimensional real vector space W . Because $A^* = -A$ and $B^* = -B$, the Hilbert-Schmidt adjoint of R is $R^* = -R$. Thus

$$\operatorname{im} R = (\ker R^*)^\perp = (\ker R)^\perp.$$

We now compute $\ker R$. After complexifying, the eigenvalues of A are $0, \pm i\alpha_1, \dots, \pm i\alpha_m$, and the eigenvalues of B are $0, \pm i\beta_1, \dots, \pm i\beta_m$. On a rank-one map carrying a B -eigenvector with eigenvalue μ to an A -eigenvector with eigenvalue λ , the complexified Sylvester map has eigenvalue $\lambda + \mu$. Since the nonzero frequency sets are disjoint, the only way to have $\lambda + \mu = 0$ is $\lambda = \mu = 0$. Hence

$$\ker R = \mathbb{R}K, \quad K(h) = \langle h, b \rangle a.$$

The Hilbert-Schmidt orthogonality condition for C is

$$\langle C, K \rangle_{\text{HS}} = \operatorname{tr}(K^*C) = \langle Cb, a \rangle = 0.$$

Therefore $C \in (\ker R)^\perp = \operatorname{im} R$. Since $C \in W$ was arbitrary, every element of W lies in the range of some zero-numerical-radius operator on W . Proposition 3.10(c) of [1] now yields $n_G(W, Y) = 0$ for every Banach space Y and every norm-one $G \in \mathcal{L}(W, Y)$. $\square$

Verification Notes

The proof above is exact finite-dimensional linear algebra. The included script `code/verify_sylvester_cover.py` only sanity-checks the Sylvester range computation numerically. The command

```
cd runs/fa_banach_001/solutions/full/\
1905.12257_odd_dim_operator_space_zero_index
conda run --no-capture-output -n sandbox python \
code/verify_sylvester_cover.py --dims 3 5 7 --trials 20
```

returned ranks 8, 24, 48, i.e. $n^2 - 1$, with residuals below $1.2 \cdot 10^{-13}$. This is consistent with the proof that the only obstruction is the line spanned by $a \otimes b^*$, but the computation is not used as evidence in place of the proof.

Novelty and Scope

Bounded local and web checks were performed on 2026-06-26. The local run indexes were searched for arXiv:1905.12257, “numerical index with respect”, “odd dimension”, and the operator-space phrasing. Web/arXiv searches used the exact phrases “When H has odd dimension”, “numerical index with respect to an operator” with “odd dimension” and “ $L(H)$ ”, the source title with “odd dimension”, and the displayed expression $n_G(\mathcal{L}(H), Y) = 0$. No later paper or existing packet answering this exact question was found; the only direct web hit was the source arXiv record.

The packet answers only the odd-dimensional Hilbert-operator-space question after Theorem 4.7. The other questions in the same source paper, such as the countability questions for $\mathcal{N}(Z)$, are not addressed here.

References

- [1] V. Kadets, M. Martin, J. Meri, A. Perez, and A. Quero, *On the numerical index with respect to an operator*, arXiv:1905.12257, 2019.